INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

15 September 2025

Subject SP1 – Health and Care Specialist Principles

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

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1 Company A is a health and care insurance company operating in Country X. Country Y is a neighbouring country to Country X with very similar but not identical language and culture as Country X. Company A's management is proposing to enter Country Y's insurance market by setting up an insurance company in Country Y. This insurance company will specialise in selling Critical Illness (CI) products to group schemes sponsored by Country Y's employers. Company A is carrying out preliminary investigations into whether Country Y will be a suitable market to enter.

(i) Describe the main features of a group CI product. [3]

(ii) Suggest possible reasons why Company A would want to set up an insurance company in Country Y and specialise in selling group CI products. [5]

(iii) Discuss the factors that Company A should consider before deciding whether to proceed with the proposal to set up an insurance company in Country Y. [10]

Company A's preliminary investigations identified the following facts:

- The overall (i.e. all critical illnesses combined) morbidity experience over the last 20 years had been lower in Country Y than Country X.
- Historical inflation rates had been much lower in Country Y than Country X in the last 5 years.
- The average salary in Country Y is marginally higher than Country X.

(iv) Discuss how Company A may make use of these facts in the following processes for the group CI products to be sold in Country Y:

(a) product design

(b) underwriting

(c) pricing.

[6]
[Total 24]

2 Company B is a large health and care insurance company in Country Z. It offers a full range of health and care insurance products and is a market leader in Country Z's Income Protection (IP) insurance market.

(i) Describe briefly the following product features in the context of IP products:

(a) rehabilitation benefits

(b) linked claims period.

[4]

(ii) Set out the approach and process, including methodology and assumptions, which Company B could take to calculate the claims in payment reserves for its IP products. [8]

Company B's management is concerned that the company may be exposed to significant counterparty risks.

(iii) Discuss the key risks to which Company B may be exposed in respect of:

(a) financial intermediaries (brokers).

(b) third party administration providers.

(c) reinsurers.

(d) professional advisers.

[8]

(iv) Suggest measures that the company could adopt in order to mitigate and manage each of the risks identified in part (iii). [6]

[Total 26]

3 Company C is a large health and care insurance company that sells Long Term Care Insurance (LTCI) in Country X. The company sells both pre-funded and immediate needs versions of LTCI.

(i) Describe briefly the differences between the main product features of pre-funded LTCI and immediate needs LTCI. [3]

The results of Company C's recent experience investigation show that over the last 5 years the pre-funded LTCI product sold lower levels of new business than the immediate needs LTCI product.

(ii) Suggest possible reasons for this experience. [5]

In a recent budget statement, the Minister of Finance of Country X highlighted the following expectations for the government's 5-year outlook for Country X:

- Annual gross domestic product growth is expected to be significantly lower over the next 5 years compared to the last 5 years.
- Annual inflation is expected to be significantly higher over the next 5 years compared to the last 5 years.
- The government expects to reduce direct spending on social services for the elderly and instead, promote private financing of health and care insurance for the elderly.
- The average life expectancy is expected to increase over the next 5 years from 75 years to at least 80 years.
- The government will investigate increasing the state pension age for citizens of Country X.

(iii) Discuss the opportunities and risks to Company C of each of the above expectations. [12]

(iv) Suggest possible actions that Company C could take to mitigate the risks identified in part (iii). [8]

[Total 28]

4 Company D is a large health and care insurance company in Country Y, specialising in Private Medical Insurance (PMI) products. Company D's marketing director has proposed that a new PMI product is designed specifically to meet the needs of Country Y's athletes across field, track and long-distance disciplines. The marketing director's proposal is based on the following observations:

- Country Y's recent success in winning gold medals by local athletes in international events including the Olympics and the World Championships has increased the number of citizens aspiring to pursue a career in athletics.
- Country Y's existing PMI products are not tailored for the specific needs of athletes because of the higher level of injuries requiring specialist treatment and the need for medical cover at overseas events where products have territorial limits.
- Athletes are role models in society and can be used by Company D for marketing purposes to enhance its public image.

Discuss the factors that Company D should consider before agreeing to the marketing director's proposal. [12]

5 Company E is a health and care insurance company that sells a full range of health and care products in Country Z.

Country Z's current regulatory regime requires health and care insurance companies to calculate their statutory reserves using prudent assumptions (best estimate assumptions plus a margin for prudence), plus additional capital requirements using simple formula based on written premiums.

Country Z's regulatory regime is currently being changed. The proposed changes include:

- The reserves will be calculated using best estimate assumptions.
- A separate risk margin will be calculated instead of a margin for prudence.
  - The risk margin will be calculated using the 'cost of capital' approach on top of the best estimate reserves.
  - This involves first projecting forward the future capital that the company is required to hold in respect of these risks, where this is determined at the end of each projection period (e.g. year) during the run-off of the business.
  - The projected capital is determined according to the relevant regulatory basis and is the amount required to be held in excess of the projected liabilities at each period.
  - These projected capital amounts are then multiplied by a cost of capital rate.
- In addition to the best estimate reserves and risk margin, additional solvency capital requirements will be required.
  - The level of solvency capital requirements will be based on either a risk measure approach such as 'value at risk' where the insurers will apply their own judgement when setting the parameters, or a standard formula approach where Country Z's regulator will prescribe the methodology and parameters to be used by the insurers.

Discuss the factors Company E should take into consideration in assessing the potential implications of Country Z's regulatory changes. [10]

END OF PAPER